

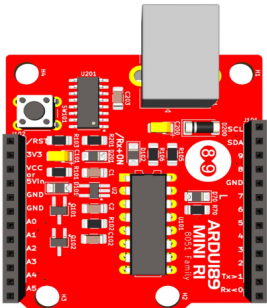
DATASHEET / USER MANUAL / ARDUI89 FAMILY
(version 2026.05)

Ardui89 MINI

8051 Family

Based on STC15W408AS-DIP16

8051 Ardui89



Ardui89 8051

/RST	
3V3	
VCC	
or 5Vin	
GND	
GND	
P1.0	A0/8
P1.1	A1/9
P1.2	A2
P1.3	A3
P1.4	A4/SDA
P1.5	A5/SCL

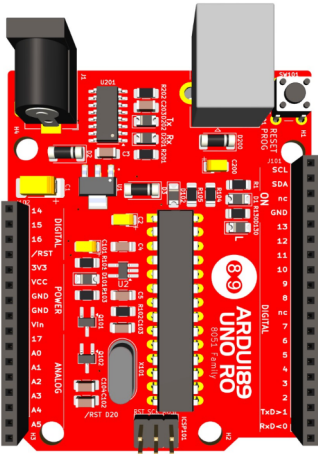
A5/SCL	P1.5
A4/SDA	P1.4
A1/9	P1.1
A0/8	P1.0
GND	
7	P3.7
6	P3.6
5	P5.5
4	P5.4
3	P3.3
2	P3.2
1/TxD	P3.1
0/RxD	P3.0

Ardui89 UNO

8051 Family

Based on STC15F2K16 (or 32)

8051 Ardui89



Ardui89 8051

P2.5	14
P2.6	15
P2.7	16+RST
	3V3
	VCC
	GND
	GND
	Vin
	17
P1.0	A0
P1.1	A1
P1.2	A2
P1.3	A3
P1.4	A4/SDA
P1.5	A5/SCL

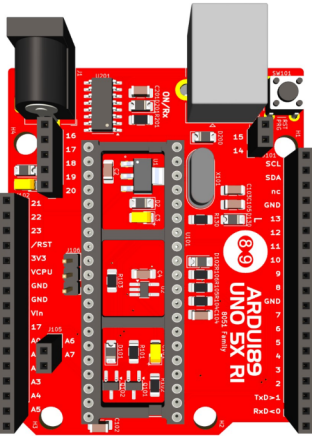
A5/SCL	P1.5
A4/SDA	P1.4
nc	
GND	
13	P2.5
12	P2.4
11	P2.3
10	P2.2
9	P2.1
8	P2.0
7	
6	P3.6
5	P3.5
4	P3.4
3	P3.3
2	P3.2
1/TxD	P3.1
0/RxD	P3.0

Ardui89 Uno 5X

8051 Family

Based on 8051 / 89C5x / 89C51x

8051 Ardui89



Ardui89 8051

P2.0	16
P2.1	17
P2.2	18
P2.3	19
P2.4	20
P2.5	21
P2.6	22
P2.7	23
	/RST
	3V3
	VCC
	GND
	GND
	Vin
	RESET
P1.7 P0.2	A0 A6
P1.6 P0.1	A1 A7
P0.5	A2
P0.4	A3
P0.3	A4/SDA
P0.2	A5/SCL

15	P3.7
14	P3.6
A5/SCL	P0.2
A4/SDA	P0.3
nc	
GND	
13	P3.5
12	P3.4
11	P3.3
10	P3.2
9	P1.1
8	P1.0
nc	
7	P1.7
6	P1.6
5	P1.5
4	P1.4
3	P1.3
2	P1.2
1/TxD	P3.1
0/RxD	P3.0

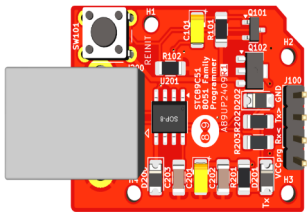
Ardui89 USB Universal Programmer

8051 Family

Self-Programmer for most STC micro-controllers
(STC89C5x/51x, STC15W..., STC32...)

PC

USB B



Ardui89

GND
Tx>
Rx<
VCCprog

Arduino IDE®

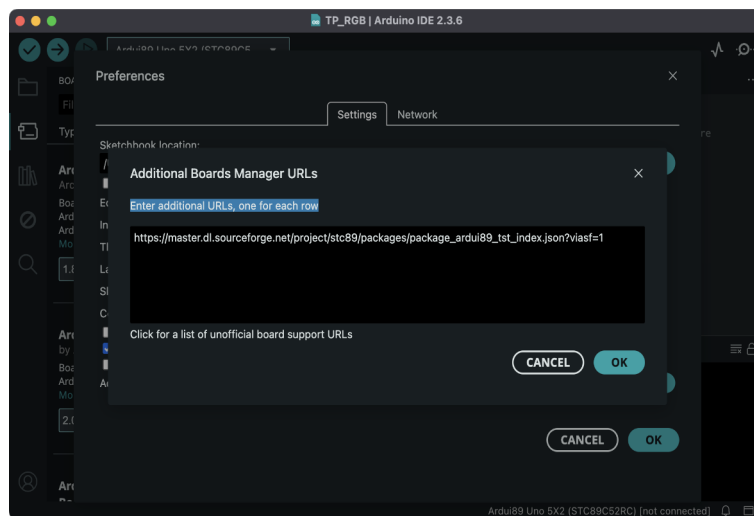
1/ Download Arduino >= V1.8.19

2/ Launch Arduino

3/ In Arduino->Preferences->Additional Board Manager URLs

add for Arduino All Versions :

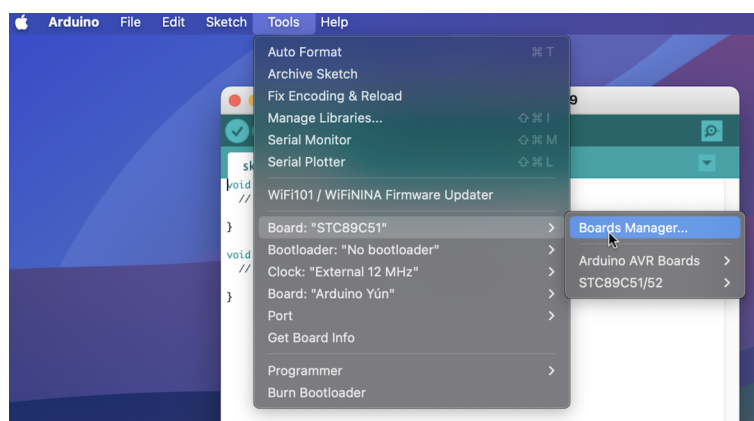
https://master.dl.sourceforge.net/project/stc89/packages/package_ardui89_index.json?viasf=1



4/ In Tools->Boards:...->Board Manager

search "8051"

select and install "ardui89 8051 family Boards"



5/ In Tools->Board:...

select "Board : "STC89C51" "STC89C52" "....""

6/ Demos :

Use Blink.ino in basics examples or in Ardui89_demos path :

Hello.ino

Howareyou.ino

and build and upload !

Examples



Functions Arduino / Ardui89

analogPin(pin) : since 0.4.2, optimized code that saves 200 bytes, configure a pin as analog input (A0, A1..)

or **pinMode(pin, INPUT_ANALOG)** : since version 0.4.2, Arduino style code, configure a pin as an analog input (A0, A1..) but with 200 more bytes.

analogRead(pin) : read analog value from analog pin channel (A0, A1...)

digitalWrite(pin, state) : since 0.0.1

digitalRead(pin) : since 0.3.7

delay(ms) : since 0.0.1

delayMicroseconds(us) : since 0.0.1

int_to_string(int) : since 0.3.10, exclusively in C for 8051

memcpy_P(dest, src, size) : since 0.3.10

pinMode : since 0.0.1

PROMEM : since 0.0.1

8051/prgspace.h : since 0.3.10

Serial89Init() : since version 0.0.1, exclusively in C for the 8051 architecture. Required for use of the serial connection.

Example :

```
// Serial port init
#if defined(ARDUINO_ARCH_8051)
Serial89Init(); // Serial 89 init
#endif
```

```
Serial.begin(9600); // Monitor init (F_CPU = 11.0592Mhz)
```

```
// Message
Serial.println("Hello, the world !");
```

Serial.begin(bauds) : since 0.0.1

Serial.print(data) : since 0.0.1 like printf in C, not Arduino C++

Serial.println(data) : since 0.0.1 like printf in C, with \n, not Arduino C++

tone(frequency (, duration)) : since 0.0.1